Harsath Kaliappa Gounder Thangavel

+91 8072415729 | harsath@fastmail.com | github.com/harsath | linkedin.com/in/harsath | harsath.com

Software engineer with 1.5 years of experience specializing in backend development. Actively contributing to open source since 2017, proficient in Python, cloud computing, and managing high-performance systems in Linux environments.

Education

York University — Toronto, Canada
Bachelor of Science (BSc) in Computer Science (Honours) — January 2020 – December 2024

Professional Experience

Software Developer — October 2023 – January 2024
Privex Inc (privex.io) — London, United Kingdom (remote)

- Developed an AI-driven customer support assistant using Retrieval-Augmented Generation (RAG) on a self-hosted LLM, delivering personalized responses to 800+ customers about support tickets, products, and services.
- Reduced support staff workload by 80% while providing instantaneous, context-aware assistance that improved overall customer experience.
- Implemented REST APIs to interface with the LLM and a Redis-based asynchronous task queue for processing customer queries and staff updates via Discord, prioritizing high throughput and robustness under heavy load.
- Designed data ingestion workflows across multiple channels, including support tickets, website documentation, and Discord announcements.
- Built the infrastructure using MySQL for chunked data storage and ChromaDB for embeddings.
- **Tech Stack:** Python (Flask, Chroma, Celery, Langchain), Redis, MySQL, Docker, Linux

Machine Learning Engineer — April 2019 – December 2019
CogniPet AG — Zurich, Switzerland (remote)

- Developed a Pet Breed Detection system using Xception neural network and transfer learning, achieving 94% accuracy across 120 dog breeds.
- Implemented REST APIs for model inference and integrated them into CogniPet's Android and iOS apps using AWS EC2.
- Built data management workflows for user-uploaded pet images in AWS S3 with GDPR compliance (timed deletion + transparency).
- Explored video-based techniques for pet facial recognition through multi-frame feature extraction analysis.
- **Tech Stack:** Python (Flask, Keras, Pandas, NumPy), AWS (EC2, S3)

Machine Learning Consultant — May 2019 – August 2019
Gnanamani Ventures Pvt. Ltd. — Namakkal, India (on-site)

- Researched and developed a facial recognition-based attendance system using the FaceNet architecture on NVIDIA Jetson Nano, optimizing for edge computing performance.
- Developed data pipelines to pre-process and fine-tune video data of 30+ staff members at Gnanamani College of Technology, resulting in a successful MVP that automated attendance tracking.
- **Tech Stack:** Python (TensorFlow, NumPy), NVIDIA cuDNN, OpenCV, Microsoft Excel

Projects

WikiScroll.xyz | JavaScript, React, Django, Docker, PostgresSQL

- Built a TikTok-style app for exploring Wikipedia articles with infinite scrolling, bookmarking, and topic-based feeds.
- Used React for frontend, Django for backend REST APIs, and PostgresSQL for data storage.

Jix Interpreter | C, CMake, GitHub Actions

- Developed an interpreter for a dynamically typed programming language in C.
- Implemented fault-tolerant parsing and error handling, arrays, function pointers, loops, nested blocks, recursion, built-in functions, conditional statements, and control flow statements.
- Implemented functional programming support, including first-class and higher-order functions.

Prism Coverage | Java, Maven, JUnit, GitHub Actions

- Developed a Java test code coverage engine to measure test suite coverage against control flow and data-flow criteria.
- Implemented color-coded HTML report generation highlighting covered and uncovered sections of source code.
- Built JUnit test suite and GitHub Actions CI/CD workflow to verify coverage engine functionality.

Technical Skills

Languages:	Python, Java, JavaScript (ES6), TypeScript, C++ (C++17), C, Bash, HTML5/CSS3
Technologies:	Linux, Docker, SQL (MySQL, Postgres), NoSQL (MongoDB, Redis, Neo4j), Kafka, AWS
Frameworks:	Node.js, React.js, Flask, Django, LangChain
Tools:	Git, Pandas, CI/CD (Github Actions, Jenkins), Build tools (CMake, Maven, Parcel), Jira